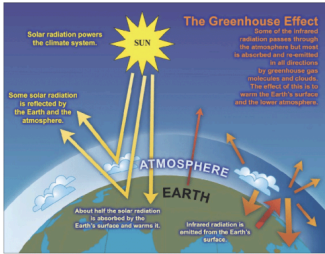


Reasoning:

Lesson 1 starts the important conversation of what a scientific model is. Oftentimes students enter a science classroom thinking that a model is a static representation of an object (i.e. cell, globe). This lesson expands that definition to an explanation of a scientific phenomenon based on evidence. Students work in groups to negotiate the evidence they collect and create a model of an observed phenomena; this highlights the idea that science is a collaborative process that requires argumentation amongst peers. This initial model-building activity activates student prior knowledge and reveals any preconceptions related to scientific models. The idea of collaboration between scientific colleagues is further amplified by the peer feedback process. Students use the observations made of the class’ models to develop a list of criteria for good models. This list of criteria will be edited as additional evidence of good models comes to light throughout the unit, highlighting the idea that scientific understanding is dynamic and can be adjusted as new evidence is revealed. Time for metacognitive reflection is incorporated at the end of the lesson to increase the likelihood of retention and transfer.

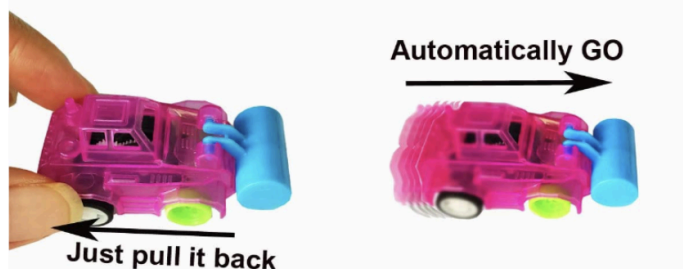
Slide	Notes
DO NOW! LESSON 1 Scientists both make and use scientific models to understand and explain the world. But how might scientists determine which models to use? Instructions: Write your thoughts on this in our "Thoughts & Reflections" sheet (in the tab with your name on it).	Follow think-pair-share protocol 💡 Elevate student responses that discuss data 💡
In this first unit, we will explore how scientists think about the world. We'll start by engaging in an important scientific practice: modeling. Models are representations that explain how the world is. Here are some examples of scientific models. 	For slides 5-7, you do not need to focus on the details of each model. Focus on the idea that these models are more than static representations of objects or equations. Each example provides an explanation for a scientific phenomenon 🌍🔍

Now you will make models in groups.

Your model should try to explain how these cars work.



Demo how the car works for your students before sending them to work in their groups



Your group has a toy car at the lab table.

Rules:

- You may not open/break the car at any time, even to "check your guess."
- You may not intentionally "break" the cars (throw them hard on the ground, etc).

Your goals:

- Gather evidence to determine how this car works
- Draw a model that represents what you conclude about the likely internal mechanism.
- Include a written explanation that explains your model and what evidence you have to support your model.



If available, have students design their models on whiteboards. This allows for easy modification. 💡

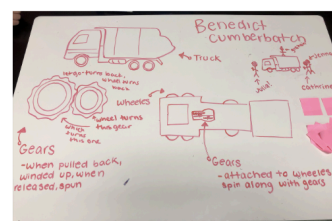
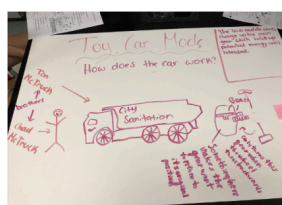
TASK EXPECTATIONS

You need to include the following parameters (and also be ready to present them to class!) in your model:

- Your visual explanation (draw it)
- The important evidence you gathered to support your model
- Your written explanation (how does your model explain/fit all of your evidence?)

Talk the class through the 3 expectations for their models. You can either do this prior to groupwork starting or pause the groups as they work with the toy cars.

EXAMPLES OF STUDENT MODELS



Some of the evidence that students may develop include:

- Car could push 5 markers but not scissors or eraser.
- Car travels only straight.
- Car moved forward with more force than pulled back
- Car moved various speeds on different surfaces
- If you wind the car on a rougher surface, it goes farther.
- If you wind the car up longer, it goes farther.

NORMS FOR SHARING AND COMPARING

1. Compare your evidence, models, and written explanation with another group.
2. Did the other group collect evidence that you didn't consider? If so, does the new evidence still fit with your model? If the new evidence does not fit with your model, think about why and how you can change your model to incorporate the new evidence.
3. Discuss what questions you have, and what other evidence you could have gathered to try to answer these questions.

Remind students that they should consider how well the models they observe fit with evidence. 

Students may need some guidance on how to provide constructive feedback to their peers. 

Possible sentence starters that can be provided to students are:

- One thing that is very clear about your model is...
- To improve on the clarity of your model, consider adding...
- It seems as though you are lacking data to support _____
- Have you considered....

NEGOTIATING THE CLASS CRITERIA FOR 'GOOD SCIENTIFIC MODELS'

Once students generate a list of criteria for good models individually, lead a class discussion to develop a class list of criteria for good models. These are criteria that the class agrees to. It's ideal to create some kind of artifact (a poster, print-out, etc.) to be on open display in the classroom. 💡

Core question: What are the characteristics—the criteria—for good models? 🔍

Elicit ideas from students. When students give their ideas, probe deeper with questions such as these, with different criteria substituted for the blanks: 💡

1. Why should good models _____ (fit the evidence, be simple, show steps, fit the evidence, etc.)?
 - a. Why is it important to put this on our list?
 - b. What do other people think about Sarah's criterion?
2. Which model shows an example of this criterion?
3. Which model shows a step by step process?
 - a. What in this model makes you think that this shows a clear step by step process?
4. What is an example of a model you looked at that has good fit with the evidence?
5. Why do you think it fits the evidence well?
6. What is an example of a model that is really clear?
7. Which of the models that you looked at was not clear?
 - a. Why do you think it was not clear.

PLEASE ENCOURAGE STUDENTS TO DO THESE TWO THINGS: 🔍

1. Explain their reasons for why particular criteria are important.
2. Explain why particular criteria are or are not found in different models that they looked at.

AVOID ONE WORD CRITERIA

SOME CRITERIA TO AIM FOR: 🔍

1. Fits with all quality evidence
2. Perspective/Size
proportion/Shading--showing multiple angles when possible
3. Include labels when drawings are unclear
4. Prioritize key attributes
5. Neatness/tell a story, so a viewer can follow
6. Don't generalize, be as exact as possible
7. Useful for making predictions

END OF LESSON 1 REFLECTION

How might scientists determine which models to use to understand something in the world?

Instructions:

Write your thoughts on this in our "Thoughts & Reflections" sheet (in the tab with your name on it).

Three core ideas that can be emphasized. You can emphasize these if students make the point, or you can introduce the point yourself if no students do it.   

1. Social interactions. Like us in this class, scientists work socially, sharing ideas, critiquing ideas, and getting better ideas from talking to each other, and learning about each other's models and evidence. At any time, some scientists may be wrong, as ideas get shared and critiqued.
2. Criteria for good models. Although scientists don't have a meeting to develop and list their criteria for good scientific models, they have a shared set of criteria very much like the criteria we have developed.
3. Models are tested by evidence; you can check their accuracy apart from their fit with evidence. In our case, we could take the car apart, and perhaps see how it actually works. But in science, scientists often cannot do this. They cannot open a cell and see what it's doing exactly. The cell is too small, and its activities are not directly visible. All scientists can do is to work to make their models fit more and more and more evidence.

Orienting to epistemic practices and the discipline:
Explain to students some similarities and differences between their practices and scientists' practices.